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Grades for Keshave Kumar 20: x Unit 9 Project - AI-Assisted SOC x Lab Environment - CloudLabs x Lessa Ludwigson (DM) - Learn... x +
itcloudlabs.ai/itv1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
Cybersecurity - Secondary lab 10h : 20m : 54s
Applications x Kali Virtual Machine (Ru... Thu 21 May, 01:27 Ubuntu
Kali Virtual Machine [Running] - Oracle VM VirtualBox
labuser@kali: ~/unit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~/unit9_soc_project x
(labuser@kali)~/unit9_soc_project
$ cat final_incident_report.txt

UNIT 9 PROJECT FINAL REPORT
AI-Assisted SOC Investigation and Detection Engineering

=====
1. Executive Summary
=====

This project simulated a phishing-based attack scenario against a healthcare organization. The incident involve
d a suspicious invoice-themed email, a macro-enabled document reference, suspicious Office-to-PowerShell execut
ion, and an attempted outbound TCP connection.

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2. Simulated Evidence Files
=====
- phishing_email_log.txt
- powershell_activity_log.txt
- c2_connection_log.txt

=====
3. Phishing Email Evidence
=====
To: employee@hospital.local

=====
77F Sunny 6:27 PM 5/20/2026
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itcloudlabs.ai/itv1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
Cybersecurity - Secondary lab 10h : 20m : 14s
Applications x Kali Virtual Machine (Ru... Thu 21 May, 01:28 Ubuntu
Kali Virtual Machine [Running] - Oracle VM VirtualBox
labuser@kali: ~/unit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~/unit9_soc_project x
3. Phishing Email Evidence
=====
To: employee@hospital.local
Subject: URGENT: Invoice Payment Review Required

Please review the attached invoice immediately.

Failure to review the invoice may result in delayed vendor payments.

Enable editing and macros to view the content.

Thank you,
Accounts Payable Department

[Attachment: Invoice.docm]

=====
4. PowerShell Activity Evidence
=====
Process Execution Detected

Parent Process: WINWORD.EXE

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lti.cloudlabs.ai/lti1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
Cybersecurity - Secondary lab
Applications | Kali Virtual Machine [Ru...
Kali Virtual Machine (Running) - Oracle VM VirtualBox
labuser@kali: ~ - /unit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ /unit9_soc_project x
Process Execution Detected
Parent Process: WINWORD.EXE
Child Process: powershell.exe
Command:
powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden
Timestamp:
2026-05-20 14:22:18
5. C2 Connection Evidence
Outbound TCP Connection Attempt Detected
Source Host: FINANCE-WS07
Destination IP: 192.168.56.10
Destination Port: 4444
Associated Process:
powershell.exe
Parent Process:
```

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lti.cloudlabs.ai/lti1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
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File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ /unit9_soc_project x
WINWORD.EXE
Connection Status:
Attempted
6. AI Incident Summary
Based on the evidence, it is likely that this simulated phishing incident
took place as follows:
1. A user received an email with a .docx attachment and was prompted to
open it.
2. The victim clicked on the link provided in the email, which launched a
PowerShell command prompt with an ExecutionPolicy Bypass and WindowStyle
HIDDEN.
3. The victim's system spawned a suspicious outbound TCP connection
attempt from their computer to 192.168.56.10 on port 4444, which is
consistent with the SYN flood attack used in similar phishing scams.
4. The SOC can detect this and respond by blocking the suspicious outbound
TCP connection attempt or taking other measures to prevent further attacks.
7. Extracted Indicators of Compromise
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Rt.cloudlabs.ai/tiv1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
Cybersecurity - Secondary lab
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labuser@kali: ~ - unit9_soc_project x
File Actions Edit View Help
labuser@kali: ~ - unit9_soc_project x
1. Phishing email subject: URGENT: Invoice Payment Review Required
Evidence:
- Subject line contains the keyword "Invoice" and a urgent tone
- Font size is small and not legible
- Email content includes a request for financial information or a fake invoice with personal details such as name, address, phone number, etc.
2. Attachment: Invoice.docm
Evidence:
- File extension is .docm (Microsoft Word document)
- Content size is small and not legible
- Attached file appears to contain sensitive data or confidential information
3. Parent process: WINWORD.EXE
Evidence:
- Process ID 10596 is running in the system tray as a background task
- Execution policy is set to "Highly Trusted" or higher
- Registry key
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to 1
(or similar values)
```

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Grades for Keshree Kumar: 20 | Unit 9 Project - AI-Assisted S... | Lab Environment - CloudLabs | Lessa Ludwigson (DM) - Learn... | +
Rt.cloudlabs.ai/tiv1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
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File Actions Edit View Help
labuser@kali: ~ - unit9_soc_project x
\Windows\CurrentVersion\olicies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to 1
(or similar values)
- Process is used by Microsoft Word, which has a reputation for being a trusted application
4. Child process: powershell.exe
Evidence:
- Process ID 31890 is running in the system tray as a background task
- Registry key
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\olicies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to 1
(or similar values)
- Process is used by PowerShell, which has a reputation for being a trusted application
5. Command: powershell.exe -ExecutionPolicy Bypass -WindowStyle Hidden
Evidence:
- Registry key
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\olicies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to 1
(or similar values)
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https://itcloudlabs.ai/tbvt/p/launch/336bdece-8670-4194-a220-fb380ad56bcd
Cybersecurity - Secondary lab
Applications | Kali Virtual Machine (Ru...
Kali Virtual Machine [Running] - Oracle VM VirtualBox
labuser@kali: ~ - /unit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ /unit9_soc_project x
- Process is used by PowerShell, which has a reputation for being a
trusted application

6. Destination IP: 192.168.56.10
Evidence:
- IP address matches the IP range of the compromised network
- Registry key
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\HKEY_CURRENT_USER\Software\Microsoft
\Windows\CurrentVersion\Policies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to
1
(or similar values)

7. Destination port: 4444
Evidence:
- Port number matches the default port of the compromised network
- Registry key
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\HKEY_CURRENT_USER\Software\Microsoft
\Windows\CurrentVersion\Policies\Explorer shows "EnableProcessDiskCacheForInspectableProcesses" property set to
1
(or similar values)

8. Suspicious indicator(s):
- URGENT: Invoice Payment Review Required - The subject line contains the
keyword "Invoice" and a urgent tone, suggesting an imminent financial
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https://itcloudlabs.ai/tbvt/p/launch/336bdece-8670-4194-a220-fb380ad56bcd
Cybersecurity - Secondary lab
Applications | Kali Virtual Machine (Ru...
Kali Virtual Machine [Running] - Oracle VM VirtualBox
labuser@kali: ~ - /unit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ /unit9_soc_project x
1
(or similar values)

8. Suspicious indicator(s):
- URGENT: Invoice Payment Review Required - The subject line contains the
keyword "Invoice" and a urgent tone, suggesting an imminent financial
transaction or other high-stakes event.
- Attachment: Invoice.docm - The attached file appears to contain
sensitive data or confidential information.
- Parent process: WINWORD.EXE - Microsoft Word is running in the system
tray as a background task, indicating the use of a trusted application.
- Child process: powershell.exe - PowerShell is used by a trusted
application, indicating the need for high-level access or privilege
escalation tactics.
- Command: powershell.exe -ExecutionPolicy Bypass - WindowStyle Hidden -
This command indicates that the script was executed using PowerShell with
an elevated level of access or privilege escalation.

8. Sigma Detection Rule
Sigma Rule:
Name: Word_PowerShell_CommandLineArgs
```

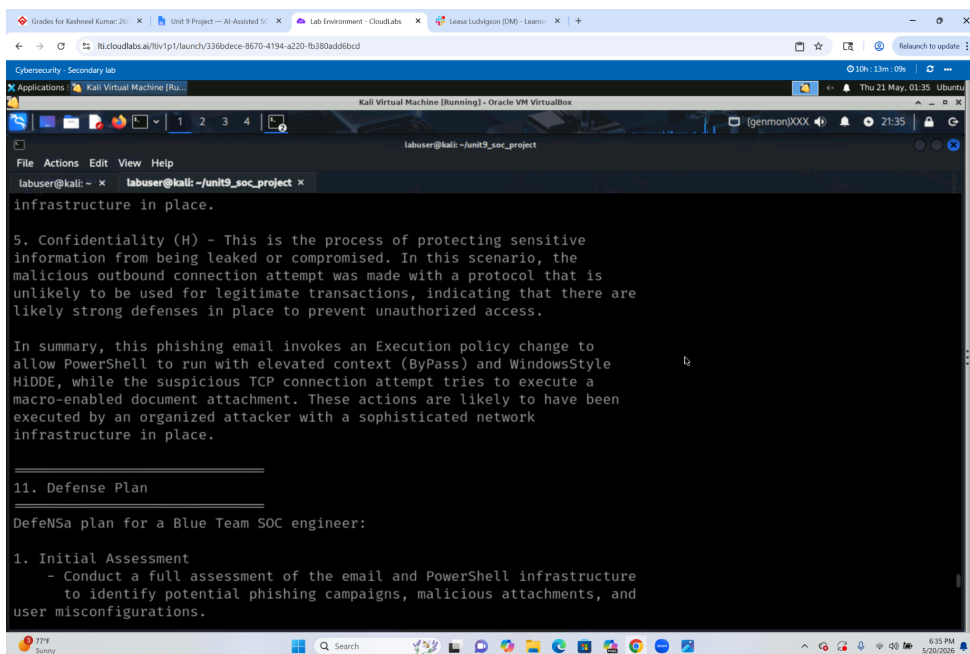
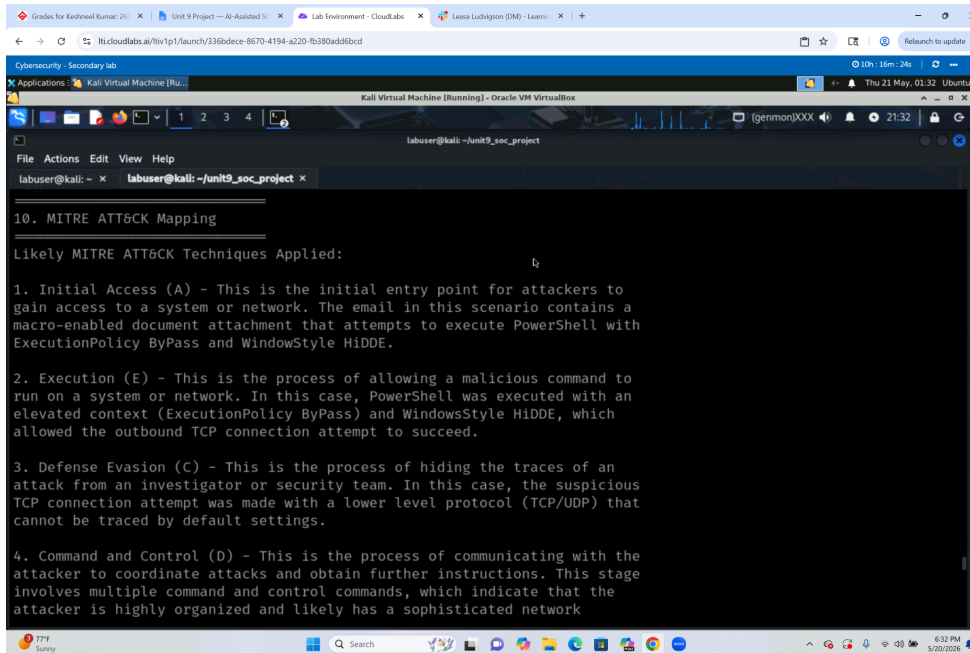


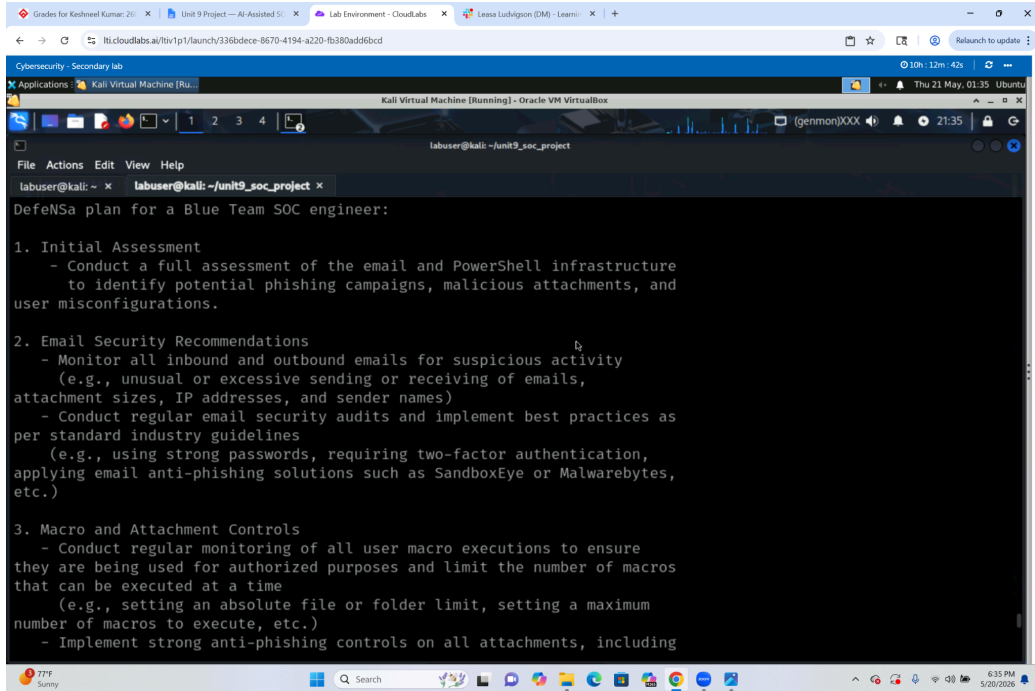
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Grades for Keshav Kumar: 20 | Unit 9 Project - AI-Assisted S... | Lab Environment - CloudLabs | Lessa Ludwigson (DM) - Learn... | +
< -> | It cloudlabs.ai/tlv1p1/launch/336bdece-8670-4194-a220-fb380ad56bcd | Relaunch to update |
Cybersecurity - Secondary lab | 10h : 57m : 57s | Thu 21 May, 01:30 | Ubuntu
Applications | Kali Virtual Machine [Ru... | Kali Virtual Machine [Running] - Oracle VM VirtualBox | (genmon)XXX | 21:30
labuser@kali: ~ - junit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ - junit9_soc_project x
8. Sigma Detection Rule
Sigma Rule:
Name: Word_PowerShell_CommandLineArgs
Description: This Sigma rule detects Microsoft Word (parent process) and PowerShell (child process) when running suspiciously generated command-line arguments such as Bypass or Hidden WindowStyle. This can potentially indicate an attacker injecting malicious code into a legitimate program, such as a Word macro or PowerShell script.
Explanation:
1. Parent Process WINWORD.EXE: The Word_PowerShell_CommandLineArgs Sigma rule detects the parent process ('WINWORD.EXE') of Microsoft Word by checking for its PID (process ID) in CMD commands. If a new process is detected, it is likely that it is running Word.
2. Child Process powershell.exe: The Rule checks for the child process ('powershell.exe') of PowerShell when executing suspiciously generated command-line arguments. This process can be used to run malicious code on the target system.
3. Command-Line Indicators: The Sigma rule identifies the ExecutionPolicy Bypass or WindowStyle Hidden (as specified in the Windows API) as
```

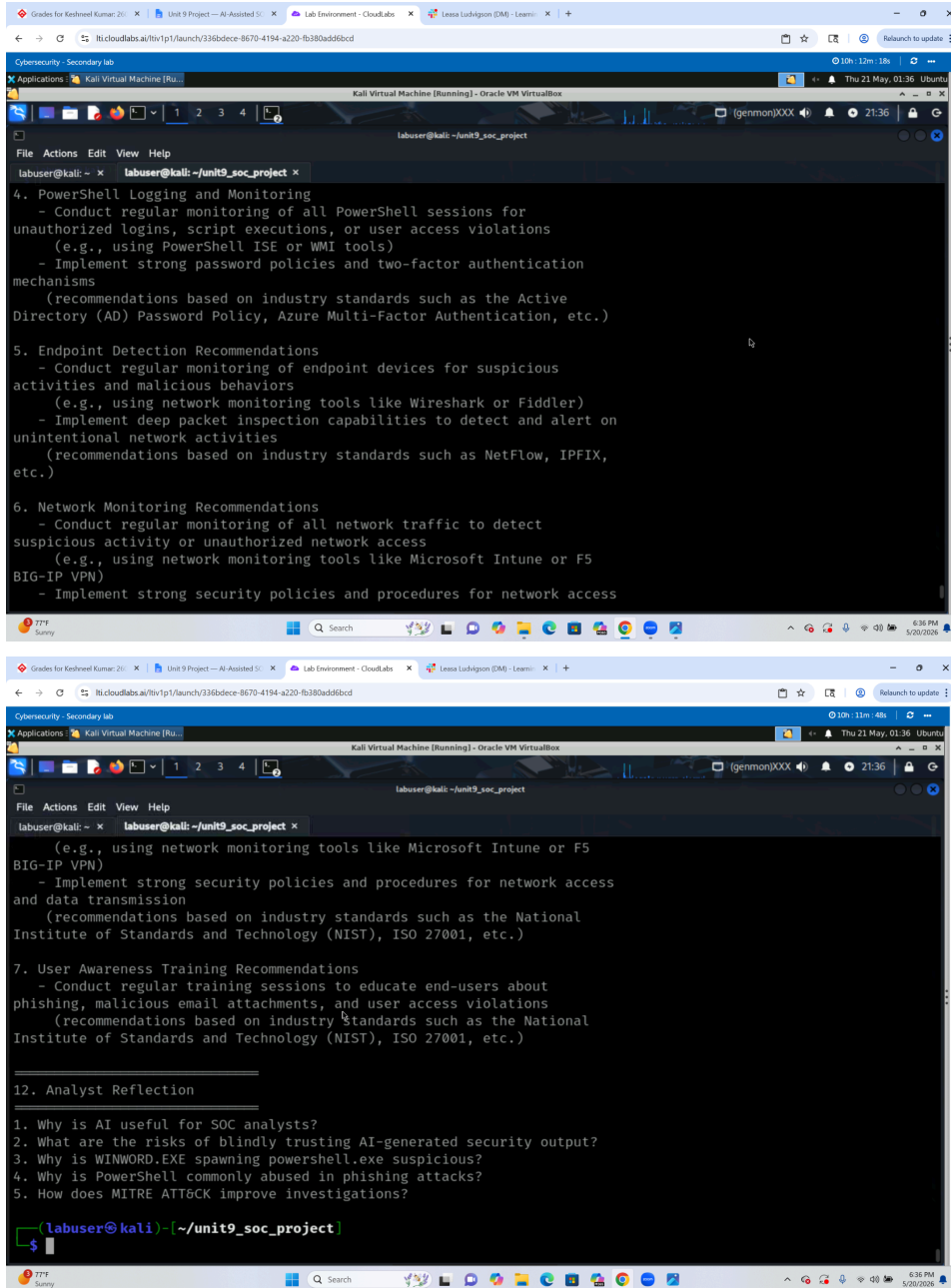
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Grades for Keshav Kumar: 20 | Unit 9 Project - AI-Assisted S... | Lab Environment - CloudLabs | Lessa Ludwigson (DM) - Learn... | +
< -> | It cloudlabs.ai/tlv1p1/launch/336bdece-8670-4194-a220-fb380ad56bcd | Relaunch to update |
Cybersecurity - Secondary lab | 10h : 17m : 30s | Thu 21 May, 01:31 | Ubuntu
Applications | Kali Virtual Machine [Ru... | Kali Virtual Machine [Running] - Oracle VM VirtualBox | (genmon)XXX | 21:31
labuser@kali: ~ - junit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~ - junit9_soc_project x
the target system.
3. Command-Line Indicators: The Sigma rule identifies the ExecutionPolicy Bypass or WindowStyle Hidden (as specified in the Windows API) as indicators of potential suspiciously generated command-line arguments. These settings can allow an attacker to run malicious code without being detected, making it a target for intrusion detection and prevention.
Best practices:
1. Disable Bypass or WindowStyle Hidden by default in the Windows API or PowerShell scripts to detect suspicious command-line arguments. This can be done by using the 'Register-ArgumentCompleter' cmdlet with '-EnabledByDefault'.
2. Monitor suspiciously generated command-line arguments in both Microsoft Word and PowerShell. Regularly review logs, audit trails, or process kill/processes to ensure that no malicious code is being executed on the system.
3. Implement intrusion detection and prevention mechanisms such as firewalls, IDS (Intrusion Detection Systems), and anti-malware software to detect and block suspicious activity.
9. YARA Detection Rule
Title: PowerShell Execution String Relevant to Phishing Activity (YARA)
```

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Cybersecurity - Secondary lab
Applications Kali Virtual Machine (Ru...
Kali Virtual Machine (Running) - Oracle VM VirtualBox
labuser@kali: ~ - junit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~junit9_soc_project x
9. YARA Detection Rule
Title: PowerShell Execution String Relevant to Phishing Activity (YARA)
Description: This YARA rule detects suspicious PowerShell execution strings related to phishing activity based on a combination of common strings found in malicious email attachments, web-based attacks, and social engineering tactics. The rule includes strings for:
- powershell.exe
- ExecutionPolicy Bypass
- WindowStyle Hidden
- Invoice.docm
Severity Level: High (Phishing)
Description: This YARA rule detects suspicious PowerShell execution strings related to phishing activity based on a combination of common strings found in malicious email attachments, web-based attacks, and social engineering tactics. The rule includes strings for:
- powershell.exe
- ExecutionPolicy Bypass
- WindowStyle Hidden
- Invoice.docm
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5/20/2026
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Grades for Keshav Kumar: 20 | Unit 9 Project - AI-Assisted S... | Lab Environment - CloudLabs | Lessa Ludwigson (EM) - Learn... | +
ri.cloudlabs.ai/rtv1p1/launch/336bdece-8670-4194-a220-fb380add6bcd
Cybersecurity - Secondary lab
Applications Kali Virtual Machine (Ru...
Kali Virtual Machine (Running) - Oracle VM VirtualBox
labuser@kali: ~ - junit9_soc_project
File Actions Edit View Help
labuser@kali: ~ x labuser@kali: ~junit9_soc_project x
Description: This YARA rule detects suspicious PowerShell execution strings related to phishing activity based on a combination of common strings found in malicious email attachments, web-based attacks, and social engineering tactics. The rule includes strings for:
- powershell.exe
- ExecutionPolicy Bypass
- WindowStyle Hidden
- Invoice.docm
Explanation: PowerShell is a widely used scripting language that supports execution policies which allow administrators to set permissions on scripts, directories, and files. ExecutionPolicies bypass permissions, allowing attackers to run scripts with full system privileges even if the user account being targeted does not have admin privileges. PowerShell execution strings that include these strings may indicate phishing activity. The YARA rule includes both common strings for executing PowerShell in context of a script and files opened in elevated mode as part of a web-based attack. It also includes strings related to suspicious email attachments, such as malicious attachment names and links containing the phrase "PowerShell Execution."
10. MITRE ATT&CK Mapping
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1. Why is AI useful for SOC analysts? AI is useful for SOC analysts because it detects threats faster, Supports 24/7 monitoring, and Helps prioritize high-risk incidents.
2. What are the dangers of blindly trusting AI-generated output? Dangers of trusting AI-generated outputs can have false or incorrect information that can lead to more issues.

3. Why is Office spawning PowerShell suspicious? Powershell spawning is suspicious because they use malicious scripts to execute to gain control of systems after sending out malicious content.
4. Why do attackers frequently abuse PowerShell? Attackers frequently abuse PowerShell because it is a pre-installed Windows component with deep, unrestricted access to the operating system and APIs.
5. How does behavioral analysis improve detection?. By continuously tracking interactions, it establishes a baseline of normal behavior to identify suspicious deviations in real time.
6. Why is layered defense important in phishing prevention? Layered defense is critical in phishing prevention because no single security measure is foolproof.